

Recent Advances in the Synthesis and Applications of Organoborane Polymers

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Abstract Recent advances in the synthesis of organoborane polymers are reviewed in this chapter. The last decade has witnessed new approaches for the direct polymerization of functional borane monomers as well as the use of innovative polymer modification techniques. Powerful new methods that allow for positioning of functional borane moieties at well-defined positions in the polymer chain have also been developed. The resulting organoborane polymers have been investigated as luminescent and electro-active materials, electrolytes for batteries, supported Lewis acid catalysts, sensors for anions and biologically relevant molecules, building blocks of stimuli-responsive and dynamic (supramolecular) materials, and for biomedical applications.

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Abbreviations

9-BBN	9-Borabicyclononane
AIBN	Azobisisobutyronitrile
Alq ₃	Aluminum tris(8-hydroxyquinolate)
Ar	Aryl
ATRP	Atom transfer radical polymerization
COD	1,5-Cyclooctadiene
Cp	Cyclopentadienyl
Cp*	Pentamethylcyclopentadienyl
CTA	Chain transfer agent
DLS	Dynamic light scattering
dppb	Bis(diphenylphosphino)butane
dppp	Bis(diphenylphosphino)propane
dtbpy	4,4'-Di- <i>t</i> -butyl-2,2'-bipyridine
Fc	Ferrocenyl or ferrocenediyl
Hex	<i>n</i> -Hexyl
LCST	Lower critical solution temperature
LLDPE	Linear low-density polyethylene
Mes	Mesityl (2,4,6-trimethylphenyl)
NIR	Near infrared
NMP	Nitroxide-mediated polymerization
OLED	Organic light emitting device
PCL	Poly(ϵ -caprolactone)
PDMA	Poly(<i>N,N</i> -dimethylacrylamide)
PEG	Polyethylene glycol
PEO	Polyethylene oxide
pin	Pinacolato
PLA	Poly(lactic acid)
PMDTA	<i>N,N,N',N'',N'''</i> -Pentamethyldiethylenetriamine
PNIPAM	Poly(<i>N</i> -isopropylacrylamide)
PS	Polystyrene
PSBA	Polystyrene boronic acid
R	Organic substituent
RAFT	Reversible addition–fragmentation chain transfer
ROMP	Ring-opening metathesis polymerization
St	Styrene or styryl
TEM	Transmission electron microscopy
<i>T</i> _g	Glass transition temperature
Tip	Tripyl (2,4,6-tri-iso-propylphenyl)

TMS	Trimethylsilyl
UCST	Upper critical solution temperature
UV	Ultraviolet

1 Introduction

Boron-containing polymers represent an emerging class of inorganic and organo-metallic polymers [1, 2]. They have long been known for their unique properties, including flame retardance and high thermal stability, and their use as preceramic materials [3]. More recently, applications as electrolytes for lithium ion batteries and as photo- and electro-active polymers (including holographic, nonlinear optical, electroluminescent materials) have been explored [4–10]. Moreover, the attachment of Lewis acidic borane groups has been exploited for the development of supported reagents and immobilized catalysts, separation media, sensor systems, stimuli-responsive materials, and biomaterials [7, 11, 12].

Organoborane-functionalized polymers are accessed either by direct polymerization of organoborane monomers or by polymer modification reactions. Both methods have been applied successfully for the attachment of new functional borane substituents to the side chains of polyolefins and conjugated polymers [3]. In addition, a diverse range of new synthetic routes for the incorporation of boron into the polymer main chain have been reported [5, 7]. In another major accomplishment, reliable methods for the synthesis of well-defined organoborane-modified polymers of controlled molecular weight and chain-end functionality have been introduced [13]. These new synthetic tools allow for the formation of more sophisticated polymer architectures such as block and star polymers and provide an entry into nanostructured boron-containing materials.

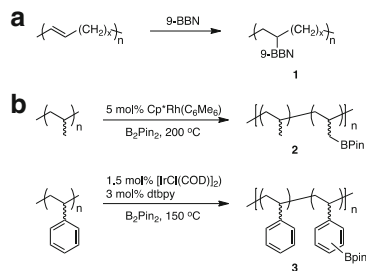
This review will highlight the recent advances in the field of boron-containing polymers with a specific focus on organoborane polymers. Polymeric materials based on carboranes are not included and those with B–N and B–P linkages [14] are discussed in a separate chapter in this book.

2 Synthesis of Organoborane Polymers

2.1 *Polymer Modification Reactions on Polyolefins*

One of the earliest and most commonly applied routes to borane-functionalized polymers involves the use of metalated polymers, especially polystyrene derivatives [3]. Suitable precursor polymers were lithiated, mercuriated, or converted to the Grignard reagent and then treated with an electrophilic borane species. However, these polymer analogous procedures often resulted in moderate chemo- and regioselectivity, and their utility was mostly limited to polymer resins [15]. In the

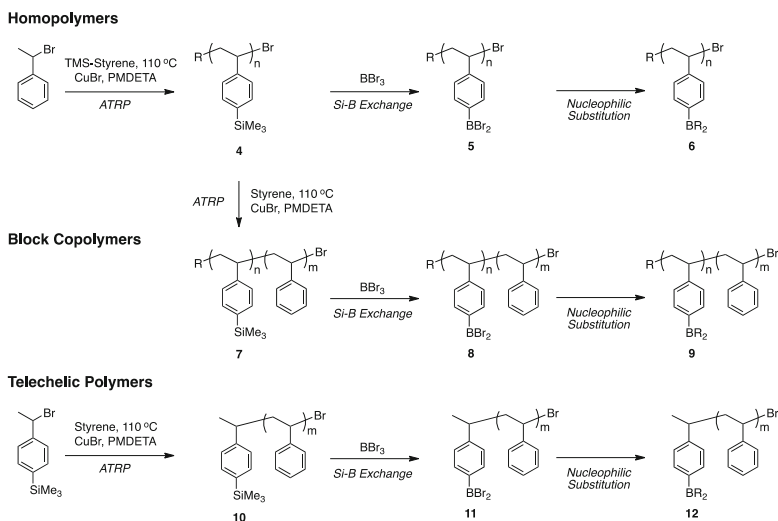
Scheme 1 Examples of polymer modification with borane groups by (a) hydroboration and (b) transition metal-catalyzed C–H activation methods



case of soluble polymers, side reactions that led to cross-linking were frequently observed [16]. Another approach entails the direct borylation of organic polymers. This method is well suited for vinyl-functionalized precursor polymers that readily undergo hydroboration (**1**, Scheme 1a) [17, 18]. However, for other less reactive polymers much harsher conditions are required; an example is the direct borylation of polystyrene with BH_2Cl , which proceeded only under forcing conditions and with low selectivity [19]. The last decade has witnessed two major advances: (a) transition metal-catalyzed direct functionalization of organic polymers using C–H activation protocols and (b) the use of highly selective Si–B exchange chemistries.

The direct borylation of polyolefins by C–H activation was first explored by Hartwig, Hillmyer, and coworkers [20–22]. They found that polypropylene can be selectively functionalized at the pendant methyl groups with B_2pin_2 (Pin = pinacolato) in the presence of $\text{Cp}^*\text{Rh}(\text{C}_6\text{Me}_6)$ as the catalyst (**2**, Scheme 1b) [21]. Similar methods were developed for the functionalization of LLDPE and for isotactic poly(1-butene) (up to 19% borylation) [22, 23]. Bae and coworkers investigated the borylation of atactic, isotactic, and syndiotactic polystyrene in the presence of an Ir catalyst (e.g., $[\text{IrCl}(\text{COD})]_2/\text{dtbpy}$) and achieved high degrees of functionalization (**3**, up to 42% borylation) [24]. Since the regioselectivity of the borylation is determined by steric effects, a mixture of *meta*- and *para*-substitution patterns was observed. The boryl moieties in the products are typically converted to organic functional groups to enhance the hydrophilicity or to allow for the formation of brush-like polymer architectures.

We established a very efficient and versatile new route to organoborane-functionalized styrenic polymers. Our approach involved the “masking” of polystyrene with silyl groups, which were then quantitatively replaced with the desired borane functional groups (Scheme 2) [25, 26]. In the first step, the preparation of the TMS-functionalized polymer **4** was accomplished by ATRP [27], a controlled free radical polymerization technique that allows us to adjust the molecular weight by simply varying the ratio of initiator to monomer. The second step involved a highly selective polymer modification reaction, where **4** was treated with a slight excess of BBr_3 to form polymer **5**. This methodology represents an exceptionally simple and straightforward route to borane-functionalized polystyrene that avoids the use of toxic mercury or tin species, while at the same time providing outstanding selectivity without any apparent cross-linking of the polymer chains. The reactive



Scheme 2 Synthetic approach to borylated homo, block, and telechelic polymers via Si–B exchange ($R=1$ -phenylethyl)

polymer **5** proved highly versatile in that the characteristics of the individual boron sites (e.g., Lewis acidity, photophysical and electronic properties) can be readily fine-tuned through high-yielding substitution reactions. A family of new organoborane polymers (**6**; $R = H$, alkyl, aryl, alkoxy, amino, etc.) was obtained, including different luminescent polymers and highly Lewis acidic triarylborane polymers that are otherwise difficult to prepare [28–32].

Another important benefit is that we were able to realize different polymer architectures. Thus, the same synthetic approach was successfully applied to the synthesis of *random copolymers*, *telechelic* (end-functionalized) polymers, and *block copolymers* containing organoborane moieties at well-defined positions of the polymer chain [33–35]. Random copolymers with both silyl and boryl groups were obtained by simply varying the ratio of BBr_3 to $SiMe_3$ functionalities in **4** [35]. Polymer **4** was also used as a macro-initiator for the polymerization of styrene [33]. The block copolymer **7** was then again treated with BBr_3 , and the resulting polymer **8** served as a “universal” precursor to amphiphilic organoborane block copolymers such as the borate and boronium-functionalized materials discussed further in Sect. 3.4. Telechelic polymers were prepared using a silylated initiator, 1-bromo-1-(4-trimethylsilylphenyl)ethane, for the polymerization of styrene, followed by exchange of the TMS end group in **10** with BBr_3 [34]. Atom transfer radical coupling (ATRC) of **10** gave the respective di-telechelic α -, ω -borane-functionalized polymers. The telechelic polymers **12** can be used, for example, in the construction of polymer assemblies as described in Sect. 3.1.

2.2 Direct Polymerization of Borane Monomers by Chain Growth Methods

The direct polymerization of borane monomers also has a long history. Early studies revolved around the free radical polymerization of styryl boronic acid derivatives for biomedical and related applications, and of vinyl-functionalized borazines and borane and carborane clusters for the preparation of preceramic polymers [3]. In recent years, significant advances have been made in that a much broader range of functional monomers can now be polymerized, including monomers with highly electron-deficient borane moieties [36–38]. In another major breakthrough, the controlled chain growth polymerization of boron-containing monomers has been popularized [13]. The latter has opened up avenues to polymers of well-defined architectures that can be used to develop new functional materials for nanotechnology and biomedical applications.

Relative to the polymer modification reactions described earlier, an advantage is that the borane monomers can be isolated and purified prior to polymerization. Moreover, the functional group tolerance is generally higher as the need for reactive borane species such as BBr_3 or transition metal catalysts can be circumvented. This allows for a greater variety of other functional comonomers to be incorporated, including acrylates, acrylamides, vinylpyridines, and ethylene oxide [39–44]. However, the polymerizability of each new monomer has to be investigated and the polymerization conditions carefully optimized.

For the preparation of block copolymers, at times it is advantageous to perform the polymerization of the boron monomer first, followed by the organic monomer. In other cases it is preferable to use the preformed organic polymer as a macro-initiator to generate the boron-containing block copolymers. Small molecule initiators that are functionalized with borane groups can be used to prepare telechelic polymers. This is schematically illustrated in Fig. 1, for the case of vinyl polymers, but similar considerations also apply to other chain growth polymerization processes such as ring-opening polymerizations.

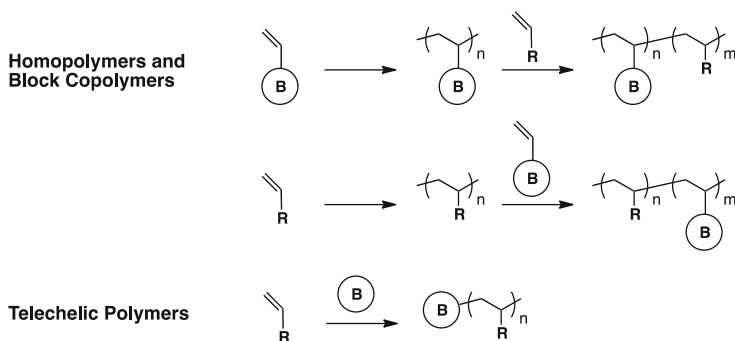


Fig. 1 Schematic illustration of the formation of homopolymers, block copolymers, and telechelic polymers from borane-functionalized vinyl monomers and initiators

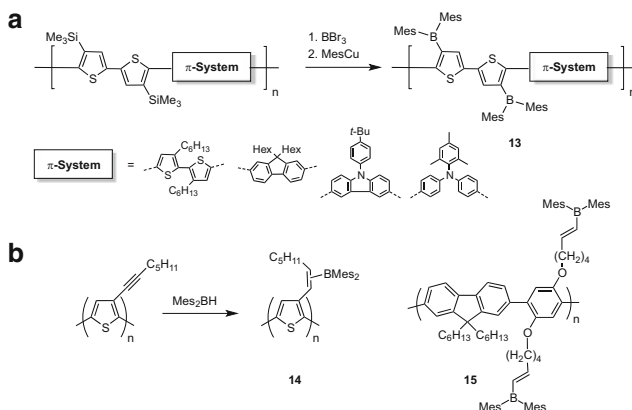
Free radical polymerizations are most frequently employed, because of the unmatched tolerance to functional groups and the typically straightforward synthetic protocols. A range of different techniques including atom transfer radical polymerization (ATRP) [27], nitroxide-mediated polymerization (NMP) [45], and reversible addition–fragmentation chain transfer polymerization (RAFT) [46] are available. It is important to emphasize that these new methods not only offer control over the molecular weight, but also provide facile access to complex polymer architectures such as block and star-shaped polymers.

We used the ATRP technique to prepare the first organoborane-functionalized block copolymers. First, the borane homopolymer (PSBpin) was obtained by ATRP of the styryl boronic ester StBpin. Subsequent chain extension with styrene proceeded with excellent efficiency and control to give the block copolymer PSBpin-*b*-PS [33]. This approach was more recently applied to other block copolymers such as PEG-*b*-PSBA (PSBA=polystyrene boronic acid) [47]. Sumerlin and coworkers first explored RAFT methods for the preparation of boronic acid-functionalized acrylamide block copolymers (see Sect. 3.3) [48, 49]. RAFT polymerizations have since proven to be highly versatile and have been applied to a wide range of different functional monomers, including (8-hydroxyquinolato) borane derivatives [43, 44, 50, 51], BODIPY-functionalized monomers [52, 53], and even Lewis acidic dimesitylborane species [37] (see Sects. 3.1 and 3.2). The NMP method was utilized in the block copolymerization of tris(2-pyridyl)borate ligand-functionalized styrene (see Sect. 3.4) [54]. Alternative approaches include Ziegler-Natta [55, 56] and ring-opening metathesis polymerization (ROMP) of borane monomers [57].

The use of borane initiators in the ring-opening polymerization of cyclic esters such as lactide and ϵ -caprolactone gives telechelic polymers. Much work has been performed on the use of boron diketonate initiators that give rise to polymeric materials with intriguing luminescence properties (see Sect. 3.2) [58–60]. Shea and coworkers explored the homologation of ylides as a means to generate borane-terminated polymethylenes [61, 62]. In another novel approach Lalevée and Lacôte used borane–carbene adducts as initiators for the radical photopolymerization of vinyl monomers [63–66].

2.3 Side-Chain Functionalization of Conjugated Polymers

Polymer modification reactions have been widely utilized for the side-chain functionalization of conjugated polymers. Typically, the goal is to modify the electronic structure of the conjugated material and thereby enhance the properties. Si–B exchange reactions are a versatile tool for the preparation of borane-functionalized polythiophenes and thiophene copolymers (Scheme 3a) [67–69]. The direct attachment of borane moieties to the conjugated backbone in polymers **13** resulted in significantly lowered LUMO orbital energies and strong bathochromic shifts in the absorption spectra. The hydroboration of poly



Scheme 3 Examples of the functionalization of conjugated polymers with electron-deficient borane moieties via (a) Si–B exchange and (b) hydroboration reactions

(3-pentylethynylthiophene) with Mes_2BH was also explored and led to a mixture of α - and β -borylated pendant groups (**14**, Scheme 3b) [70]. Fluoride anion complexation to **14** resulted in easily observed changes in the color of the materials. These changes were attributed in part to the electronic effect of converting the tricoordinate boranes to tetracoordinate borates and in part to steric effects that result in twisting of the polythiophene main chain. Another example is the introduction of electron-deficient dimesitylborane side chains into a fluorenylene-phenylene copolymer; the borylated product (**15**, Scheme 3b) was utilized for the development of p/n junctions via fluoride anion complexation [71].

Polymer modification reactions have also been proved to be tremendously useful for the attachment of boron chelate complexes to conjugated polymers [72–74]. A recent example is the complexation of fluorene-diketone copolymers with $\text{F}_3\text{B} \cdot \text{OEt}_2$ (**16**, Fig. 2a) [75]. Depending on the degree of borane complexation different luminescence colors could be realized. In a similar manner, conjugated copolymers that incorporate hydroxyquinoline groups were treated with Ph_3B (**17**, Fig. 2b) [73]. We note that this approach is not restricted to tetracoordinate boron polymers; for example, the reaction of phenylenediamine copolymers with arylboronic acids gave rise to benzodiazaborole-functionalized conjugated materials [72]. On the other hand, reaction of conjugated materials that contain Lewis basic pyridyl or thiadiazol groups with highly Lewis acidic boranes can be used to alter the electronic structure via Lewis acid–base complex formation [76].

As milder conditions for polymerization have become available and more robust monomers have been developed in recent years, direct polymerization of organoborane monomers has become increasingly attractive. In the case of conjugated materials, essentially the entire arsenal of typical C–C bond forming reactions has been exploited [77–81]. For example, Yamaguchi and coworkers prepared a series of poly(*p*-phenylene ethynylene)s (**18**) by Sonogashira–Hagihara coupling as illustrated in Scheme 4a. The modular approach allowed for incorporation of a

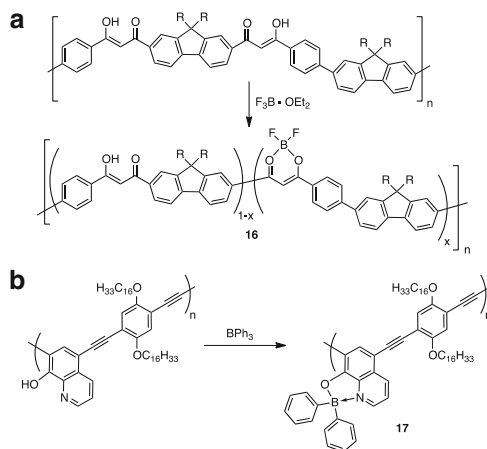
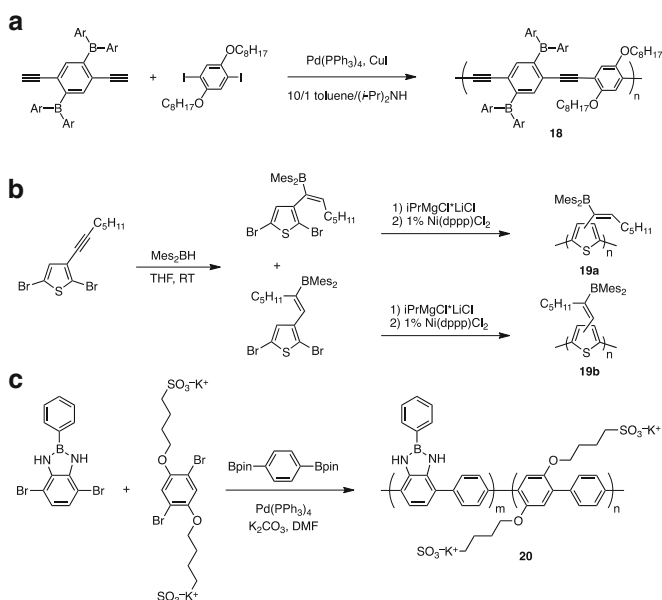


Fig. 2 (a) Functionalization of conjugated diketonate copolymers with varying amounts of $F_3B \cdot OEt_2$. (b) Functionalization of conjugated 8-hydroxyquinoline copolymer with BPh_3



Scheme 4 Examples of the synthesis of conjugated polymers with borane side groups from borane monomers via (a) Songashira-Hagihara, (b) Kumada, and (c) Suzuki-Miyaura-type polycondensation

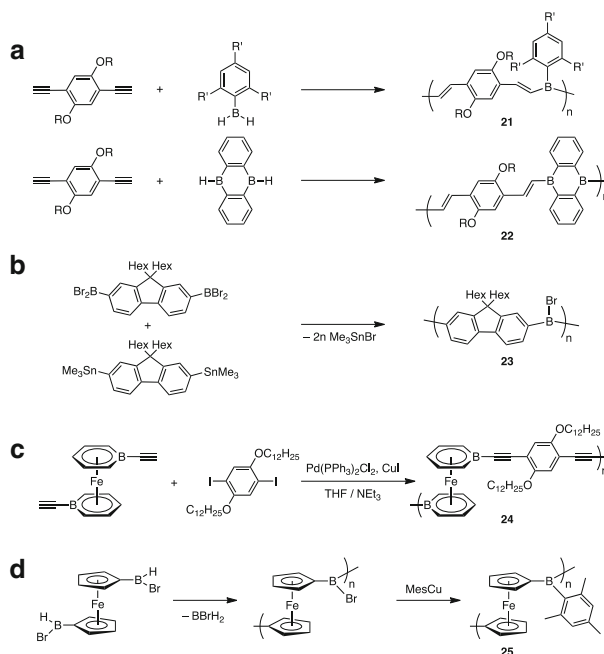
range of different diiodoarenes as comonomers [77]. We explored the Kumada coupling polymerization of thiophenes with vinylborane substituents (Scheme 4b) [78]. Hydroboration of the 3-(*n*-pentylethynyl)thiophene gave rise to two different regioisomers, which were separated by reverse phase column chromatography,

converted to the corresponding Grignard species, and then polymerized in the presence of Ni(dppp)Cl_2 . The optical properties of the polymeric products **19a** and **19b** differed significantly, highlighting the effects of the substitution pattern on the extent of cross-conjugation between polymer main chain and borane side groups. Diazaborole polymers were prepared by C–C coupling as well as electropolymerization methods [82]. For example, a water-soluble conjugated copolymer (**20**) was obtained by Suzuki–Miyaura coupling copolymerization of a dibrominated diazaborole (Scheme 4c) [83]. Finally, Chujo and coworkers incorporated BODIPY chromophores as pendant groups to polyfluorenes by Ni-catalyzed Yamamoto or Suzuki–Miyaura coupling methods [84]. The polymers exhibited interesting energy transfer from the BODIPY side chains to the polyfluorene main chain.

2.4 Boron Incorporation into the Main Chain of Conjugated Polymers

The first well-defined conjugated main chain-type polymers were prepared by Chujo and coworkers in 1998 via hydroboration polymerization (**21**, Scheme 5a) [85]. A wide range of different luminescent polymers have since been obtained by reaction of organic or organometallic dialkynes with MesBH_2 or TipBH_2 (Mes = 2,4,6-trimethylphenyl, Tip = 2,4,6-tri-*iso*-propylphenyl). The use of the bulkier Tip derivative results in enhanced stability of the products [86]. A diboraanthracene derivative with two spatially separated B–H functionalities was utilized by Wagner and coworkers to generate polymer **22** [87].

Organometallic polycondensation reactions present an attractive alternative, especially when hydroboration methods are not applicable as in poly(aryleneborane)s. Grignard and organolithium reagents were explored early on, but the molecular weights were relatively low [88]. We demonstrated more recently that reaction of distannyl species with arylboranes ArBX_2 or X(R)B-Ar-B(R)X (X = Br, Cl) results in highly selective polymer formation under mild conditions via Sn–B exchange (**23**, Scheme 5b) [89–93]. Transition metal-catalyzed C–C coupling processes have also attracted increased attention. They have been widely used to incorporate luminescent BODIPY and boron 8-hydroxyquinolate heterocycles into conjugated polymers [94–101]. In addition, Sonogashira–Hagihara coupling was used to embed bis(borabenzene)iron complexes into conjugated polymers (**24**, Scheme 5c), while a recent report details the Stille-type polymerization of thiophene derivatives of diboraanthracene [102]. Other less frequently employed techniques include the spontaneous polymerization of in situ generated ferrocenylboranes, H(Br)B-fc-B(Br)H (fc = 1,1'-ferrocenediyl) (**25**, Scheme 5d) [103], the spontaneous ring-opening polymerization of borafuorenes [104], and the use of preformed organoboranes in click reactions [105]. Electropolymerization



Scheme 5 Examples for the synthesis of main chain-type conjugated organoboron polymers via (a) hydroboration polymerization, (b) Sn–B exchange polymerization, (c) Sonogashira–Hagihara and Stille-type polycondensation, and (d) spontaneous polycondensation of arylboranes

presents yet another alternative; however, its use has been largely limited to monomers with boronic ester or carborane pendant groups.

3 Selected Examples of Polymers and Their Applications

In the following section we introduce selected classes of organoborane polymers and briefly discuss their application fields.

3.1 Borane Polymers as Lewis Acids and Anion Sensors

Among the earliest examples of polymer-supported borane Lewis acids are Merri-field resins that were functionalized with 9-BBN or Tip_2B moieties [106, 107]. These resins were used in hydroboration reactions and the reduction of ketones. More recently, we developed a new family of soluble polymers (**26**) using the Si–B exchange protocol discussed in Sect. 2.1 [108]. The Lewis acidity of the boranes in **26** was fine-tuned by modification of the substituent R (Fig. 3),



Fig. 3 Lewis acidity tuning of organoborane polymers

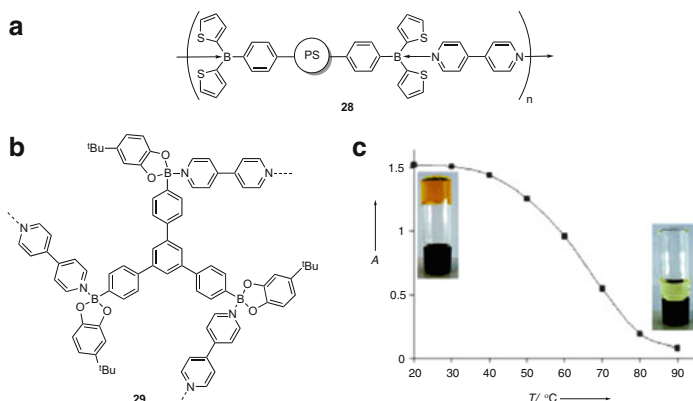
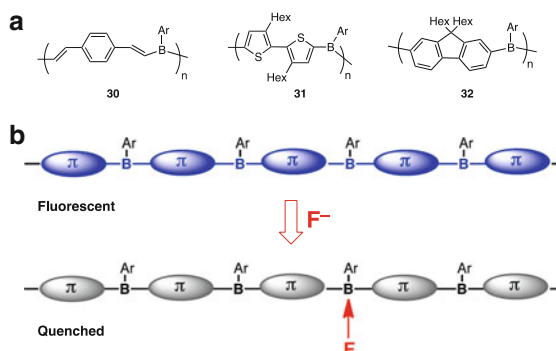


Fig. 4 (a) An example of a supramolecular 1D polymer formed via Lewis acid–base interactions (PS polystyrene). (b) Formation of a 3D network using a trifunctional boronate ester and 4,4'-bipyridine. (c) Illustration of the temperature-dependent properties of a related organogel in toluene. Adapted with permission from Sheepwash et al. [109]. Copyright 2011 Wiley-VCH Verlag GmbH & Co. KGaA

thereby offering control over the strength and reversibility of the donor–acceptor bonding to Lewis basic substrates. We found that the arylborane polymers **26** ($\text{R} = \text{C}_6\text{F}_5$, thienyl) and the related hydroborane polymer **27** strongly bind to Lewis bases as evidenced by the formation of soluble, isolable polymeric donor–acceptor complexes with pyridine ligands [28, 108]. With weaker donors, a temperature-dependent equilibrium between coordinated and uncoordinated sites was established [108]. On the other hand, more weakly Lewis acidic poly(styrene boronic esters) (**26**, $\text{R} = \text{alkoxy}$) were only partially coordinated even with very strong bases.

The self-assembly of polymeric Lewis acids and Lewis bases is expected to allow for reversible formation of complex polymer architectures such as *block* and *graft* copolymers and of reversibly cross-linked materials. An example of this concept is the assembly of borane-functionalized mono- and di-telechelic organoboron polymers into supramolecular structures by means of reversible noncovalent interactions with bifunctional donor molecules (**28**, Fig. 4a) [34]. Studies on the assembly of molecular polyfunctional organoboranes and polyfunctional amines to give 1D, 2D, and 3D-polymeric materials were also reported [109–113]. For example, Severin and coworkers discovered the formation of some very interesting temperature-responsive organogels in toluene (**29**, Fig. 4b, c) [109].

Fig. 5 (a) Examples of conjugated organoborane polymers that were studied for the recognition of anions and aromatic amines. (b) Schematic illustration of the amplified fluorescence quenching upon binding of fluoride to one of the borane sites in a conjugated organoborane polymer



At temperatures above 60°C, the organogel broke up due to rupture of the B–N bonded network and reformed again upon cooling to room temperature.

The Lewis acidity of tricoordinate organoborane polymers is also advantageous for the development of new luminescent materials for the recognition and sensing of anions. Figure 5a shows selected examples of conjugated organoborane polymers that are luminescent and at the same time act as Lewis acids [89, 90, 114]. An important feature is the overlap of the empty p-orbital of the electron-deficient boron atom with the π -conjugated organic bridges, which leads to bathochromic shifts in the absorption and frequently strong emission in the visible region. Steric shielding of the borane moiety is usually accomplished by the introduction of a bulky aryl groups (Ar = mesityl, triptyl) and results in high selectivity for small anions such as fluoride or cyanide. Coordination of these anions to boron then triggers a response in the photophysical characteristics (e.g., change in the absorption and emission color or quenching of the luminescence) [115–117], which is specific to the nature of the anion under investigation and can readily be verified via spectroscopic screening or even by naked eye observation.

An intriguing aspect of conjugated organoborane polymers is that energy transfer can occur from remote sites to the anion recognition site, which then acts as a quenching site [118]. This implies that coordination of the anion to only one of the borane moieties results in effective quenching of the fluorescence of the entire conjugated polymer chain as illustrated in Fig. 5b. This phenomenon was first qualitatively reported by Chujo and coworkers for polymer **30** [114], and a quantitative analysis of the binding of pyridine to a poly(bithiophene borane) system (**31**) was performed by our group [89]. The amplified quenching appears to be a quite general phenomenon as several other conjugated polymer systems have since been reported [90, 91, 119]. We also succeeded in the preparation of a conjugated organoborane macrocycle, which allowed for a more detailed investigation since polydispersity and end group effects do not come into play [120].

Related phenomena were also observed for nonconjugated polymers with borane chromophores as side chains [29–31, 55, 57]. As an example, polymers **33** (Fig. 6) are strongly emissive in the blue region with a maximum at $\lambda_{\text{em}} = 463$ nm when R = H, while the amino-functionalized derivative (R = NPh₂) emits in the green at

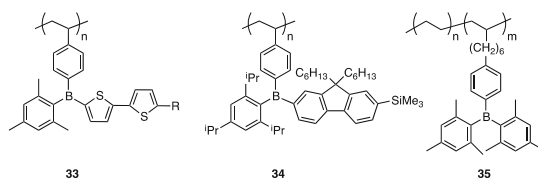


Fig. 6 Examples of polymers with fluorescent triarylborane side chains

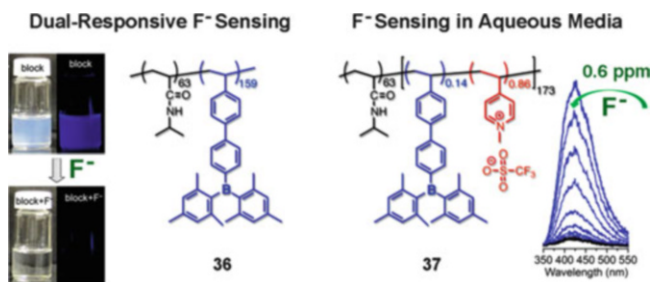


Fig. 7 Fluoride ion sensing with organoborane block copolymers. Adapted with permission from Cheng et al. [37]. Copyright 2013 American Chemical Society

$\lambda_{\text{em}} = 537 \text{ nm}$. These polymers selectively bind fluoride and cyanide over a range of other anions (e.g., Cl^- , Br^- , and NO_3^-) as reflected in a hypsochromic shift of the absorption maximum. Fluoride or cyanide binding to **33** ($\text{R} = \text{NPh}_2$) led to a change of the emission color from green to blue, but for **33** ($\text{R} = n\text{-hexyl}$) highly effective fluorescence quenching was observed. The amplified quenching effect was tentatively attributed to through-space exciton migration to lower energy fluoroborate quenching sites. Similar studies were reported by Do and Lee and coworkers for polymers **35**, which strongly emit in the violet/UV region with a maximum at 382 nm [55, 57].

We recently reported a new twist to this concept: i.e., the block copolymer **36** not only responds to fluoride anions with a color change but at the same time the solubility characteristics change dramatically (Fig. 7) [37]. In DMF solution, the borane-functionalized block forms the core of self-assembled aggregates, but upon addition of a small amount of fluoride anions the aggregates disassemble into individual polymer chains. The disassembly process can be monitored by a change in turbidity as well as the quenching of the fluorescence. We also prepared a novel block–random copolymer **37**, in which cationic pyridinium groups are inserted into the borane block. The pyridinium groups further enhance the anion binding strength due to favorable electrostatic interactions, allowing for the detection of very low levels of fluoride even in aqueous solution ($\text{DMF}/\text{H}_2\text{O} = 9:1$) [37].

3.2 Luminescent Polymers Based on Boron Chelate Complexes

Polymers that contain boron heterocycles, typically with boron in the tetracoordinate state, have been pursued vigorously due to their promise as new luminescent materials. Applications as emissive and electron-conduction materials for organic light emitting devices (OLEDs), lasing materials, in bioimaging, and as stimuli-responsive materials are most prominent. Architectural control tends to be an important aspect because it allows for the fabrication of controlled size nanoparticles, for example, by the assembly of block copolymers or star polymer.

The boron heterocycles come in many different flavors, but three classes have been studied most extensively: boron dipyrromethene (BODIPY), boron 8-hydroxyquinolate, and boron diketonate derivatives (Fig. 8). They stand out because of their high stability, typically strong emission, and facile tuning of the absorption and emission characteristics through simple substituent variations. There is a vast literature on the incorporation of these chromophores into the main chains or side chains of polymers. In the following sections, we offer a few selected examples and discuss pertinent applications where appropriate.

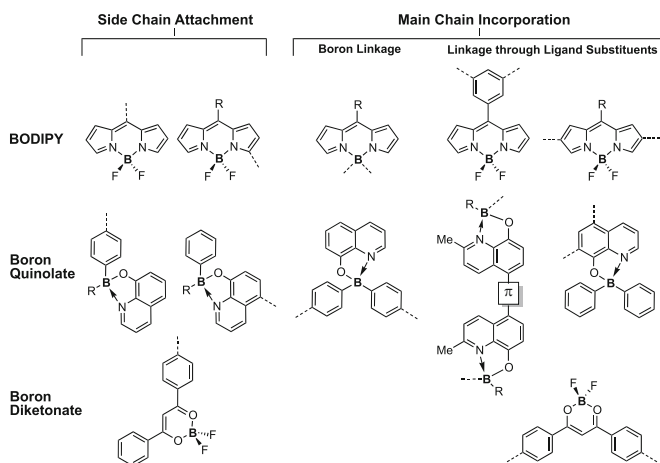
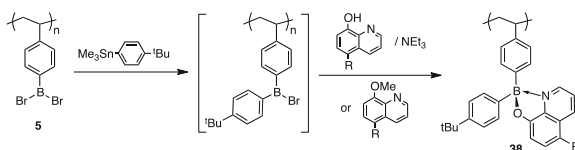


Fig. 8 Incorporation of fluorescent boron heterocycles into polymers. The linkages to the polymer chains are indicated with *dashed lines*; some of the substituents on the pyrrole and quinoline moieties are omitted for clarity reasons



Scheme 6 Synthesis of boron quinolate polymers **38**

3.2.1 Boron Quinolate-Functionalized Polymers

In 2004, we introduced the first class of polymers with boron quinolate chromophores attached to a polystyrene backbone using a facile one-pot polymer modification procedure (Scheme 6) [121–122] (see also ref. [123]). The polymeric products (**38**) proved to be highly soluble and luminescent thin films could be readily cast from solution. Importantly, we found that the emission characteristics could be tuned by simple variation of the substituent in the 5-position of the quinoline moiety ($\text{R} = \text{Bpin}$, C_6F_5 , Cl , $p\text{-C}_6\text{H}_4\text{OMe}$, $p\text{-C}_6\text{H}_4\text{NMe}_2$), covering almost the entire visible spectrum from blue to red emission [122, 124].

More recently, we explored the controlled radical polymerization of organoboron quinolate monomers in an effort at achieving architectural control. Well-defined block copolymers with polystyrene, poly(styrene-*alt*-maleic anhydride), polyethylene oxide, and poly(4-vinylpyridine) were prepared by RAFT polymerization using monomer **39** as the building block [43, 50]. A pyridine derivative of **39** ($\text{R} = 4\text{-pyridyl}$) was also prepared and incorporated into a block copolymer with PS [44]. The self-assembly of this amphiphilic block copolymer was investigated and the pyridine binding sites were exploited for the complexation to a Zn porphyrin derivative. Complexation to Zn resulted in interesting energy transfer phenomena from the boron quinolate chromophore to the porphyrin (Fig. 9a). We also prepared the first organoborane star polymers using an arm-first RAFT polymerization method, in which a distyrylboron quinolate cross-linker (**40**) was polymerized with AIBN in the presence of different macro-chain transfer agents (CTA, Fig. 9b) [51]. The successful formation of well-defined luminescent stars was confirmed by TEM and DLS analysis.

Chujo and coworkers studied the incorporation of boron quinolate chromophores into the main chain of conjugated polymers [73, 96, 125, 126]. A range of different methods were explored, including polymer modifications and the direct polymerization of preformed complexes via C–C coupling methods. For example, polymers **42** were prepared by treatment of **41** with BPh_3 or $(\text{C}_6\text{F}_5)_2\text{BF} \cdot \text{OEt}_2$ (Fig. 10) [127]. The application of **42** as an electron conducting material in OLEDs was explored. For the fluorinated polymer ($\text{Ar} = \text{C}_6\text{F}_5$), an electron mobility close to that of aluminum tris(8-hydroxyquinolate) (Alq_3) was recorded in an electron-only thin film device.

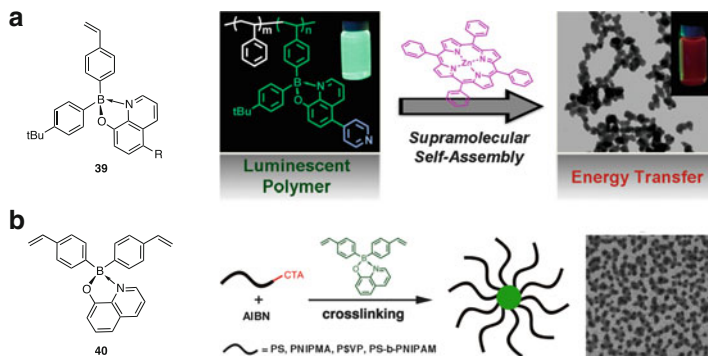


Fig. 9 (a) Structure of boron quinolate monomers (**39**) used to generate well-defined homo and block copolymers and illustration of the supramolecular assembly of a pyridyl derivative (R=4-pyridyl) with Zn tetraphenylporphyrin. Adapted with permission from Cheng et al. [44]. Copyright 2013 American Chemical Society. (b) Structure of a boron quinolate cross-linker (**40**) and its use in the formation of luminescent star polymers by arm-first RAFT polymerization. Adapted with permission from Cheng et al. [51]. Copyright 2012 The Royal Society of Chemistry

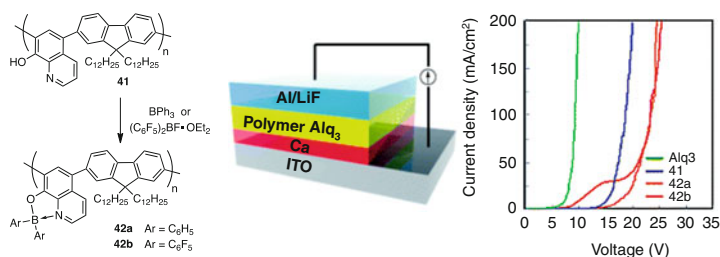


Fig. 10 Synthesis of boron quinolate polymers **42** via post-polymerization modification and current-voltage plots in comparison to an Alq_3 device. Adapted with permission from Nagai et al. [127]. Copyright 2010 American Chemical Society

3.2.2 BODIPY-Functionalized Polymers

Boron dipyrromethene dyes are among the most widely utilized fluorophores, because of their strong and sharp emission in the visible and in some cases even the NIR region. Numerous polymers have been developed with BODIPY as pendant groups or incorporated into the polymer main chain. Some examples of random and block copolymers that contain BODIPY side groups (**43–45**) are illustrated in Fig. 11 [52, 53, 128]. The first polymers with BODIPY in the main chain (**46**) were prepared by Li and Liu via Sonogashira–Hagihara coupling [94, 97]. Polymers **47** are linked through the boron atom itself; they were found to aggregate into nanosized particles and higher-order, fiber-like structures depending on the bridging aryl (Ar) moiety, possibly promoted by the presence of the long alkyl group on the dipyrromethene entity [129]. Polymers **48** are highly emissive as

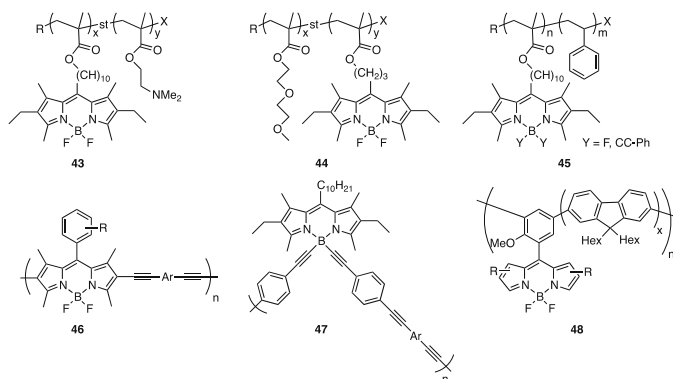


Fig. 11 Selected examples of BODIPY-functionalized polymers

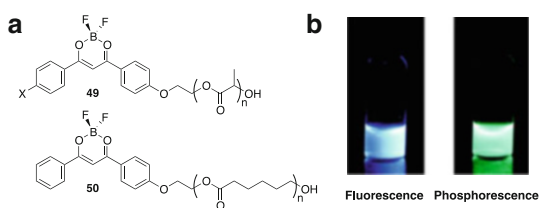


Fig. 12 (a) Boron diketonate-terminated PLA (**49**, $X=H, I$) and PCL (**50**). (b) Photographs of aqueous nanoparticle suspensions of **49** in the presence of air (fluorescence observed) and under N_2 atmosphere after turning off the excitation source (phosphorescence observed). Adapted with permission from Pfister et al. [131]. Copyright 2008 American Chemical Society

a result of efficient energy transfer from the oligofluorene bridging moieties to the BODIPY chromophores [130].

3.2.3 Boron Diketonate-Functionalized Polymers

The functionalization of polymers with boron diketonate chromophores was pioneered by Fraser and coworkers. They used boron diketonate derivatives with primary alcohol functionalities as initiators for the ring-opening polymerization of lactide and discovered that the resulting polymers (**49**, Fig. 12a) exhibited dual emission at room temperature, strong blue fluorescence, and simultaneous green phosphorescence [132]. In contrast, phosphorescence was only observed at temperatures below $0^\circ C$ for polymers (**50**) derived from ϵ -caprolactone. The different behavior was attributed to the lower glass transition temperature (T_g) of poly(ϵ -caprolactone) (PCL) in comparison to that of poly(lactic acid) (PLA) [60]. The fluorescence emission color of the PLA material was found to be also molecular weight dependent (blue to green), indicating the importance of dye–dye interactions

[133]. Nanoparticles that can be dispersed in aqueous solution and exhibit both fluorescence and room temperature phosphorescence were fabricated via nanoprecipitation from DMF as a water-miscible organic solvent [131].

An intriguing aspect of these materials is that the phosphorescence but not the fluorescence is effectively quenched by oxygen. This is illustrated for an aqueous nanoparticle suspension of **49** in Fig. 12b. In the presence of air only the blue fluorescence is apparent, while under nitrogen atmosphere the longer-lived green phosphorescence can be observed after the excitation source is turned off. This property was exploited in tumor hypoxia imaging applications [134, 135]. For this purpose a polymer with an iodine substituent (**49**, X = I) was developed which showed enhanced phosphorescence, because the iodine heavy atom promotes intersystem crossing into the triplet manifold.

As already mentioned earlier, conjugated polymers were also modified with boron diketonate functional groups (see Fig. 2a) [75].

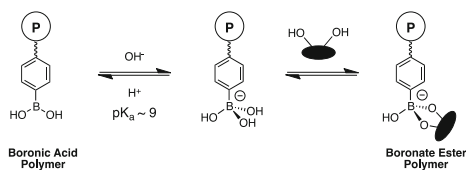
3.3 Boronic and Borinic Acid-Functionalized Polymers for Biomedical Applications and as Smart Materials

Boronic acid-functionalized polymers have garnered intense interest because of the ability of boronic acids to reversibly bind 1,2- or 1,3-diols as illustrated in Fig. 13.

This property allows for the detection and immobilization of sugars; especially the complexation of glucose is of great importance in diagnostics and treatment of diabetes. A range of other biomedical applications have been explored, and this topic is covered in a review by Sumerlin and coworkers [12]. We will focus here on recent advances in areas pertaining to (1) architectural control; (2) development of multiple responsive systems; (3) structural modification to allow for application at physiological pH; and (4) the discovery of new responsive materials.

To achieve architectural control where boronic acid sites are incorporated at well-defined positions of polymers and nanoparticles, controlled radical polymerization techniques are the method of choice. We prepared the first boronic acid block copolymer, PS-*b*-PSBA (**51**, Fig. 14), in 2005 using ATRP in combination with polymer modification procedures that involved the exchange of silyl for boryl groups [33]. The block copolymer was shown to self-assemble into micelles, vesicles, and bundles of rodlike structures, depending on the pH and the amount of THF used as an organic co-solvent [136]. Also, van Hest and Kim and coworkers

Fig. 13 Schematic illustration of the binding of sugars (1,2- or 1,3-diols) to boronic acid-functionalized polymers



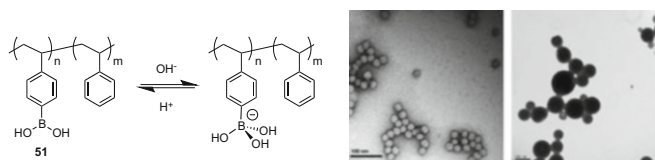


Fig. 14 Structure of PS-*b*-PSBA block copolymers (**51**) and TEM images of assemblies in 0.1 M NaOH (*left*) and in THF/H₂O mixture (*right*). Adapted with permission from Cui et al. [136]. Copyright 2010 Wiley-VCH Verlag GmbH & Co. KGaA

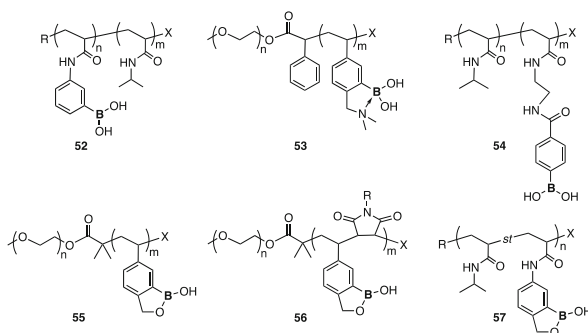


Fig. 15 Examples of boronic acid and boroxole-functionalized polymers that display dual responsive properties and/or are capable of binding sugars at physiological pH

utilized similar methods for the preparation of vesicles that consist of PEG-*b*-PSBA and PEG-*b*-PS and studied them as stimuli-responsive drug delivery vehicles [47].

Sumerlin and coworkers developed RAFT polymerization methods for the preparation of thermoresponsive acrylamide block copolymers [41, 48, 49, 137]. As an example, the acrylamide block copolymer **52** (Fig. 15) forms micellar aggregates in aqueous solution at low pH [137]. These aggregates break up when the pH is increased and sugar molecules are added to the solution. A secondary response is observed when the solution is heated above the lower critical solution temperature (LCST) of the PNIPAM block, which results in the formation of reverse micellar aggregates with PNIPAM in the core.

The sugar binding process is strongly pH dependent as the binding equilibrium relates to the pK_a of the boronic acid moiety. For practical applications it is important to achieve sugar binding at physiological pH. Several approaches have been taken toward this goal: van Hest, Kim, and coworkers used an amino-functionalized Wulff-type phenylboronic acid derivative (**53**), which favors sugar binding due to an intramolecular B–N interaction [138]. Sumerlin and coworkers demonstrated that linkage of the phenylboronic acid moiety through the electron-withdrawing carboxyl rather than amino group of an amide (**54**) results in similarly enhanced sugar binding at neutral pH [41]. In a different approach, Kiser and coworkers incorporated more acidic boroxoles [139, 140] into polymer structures

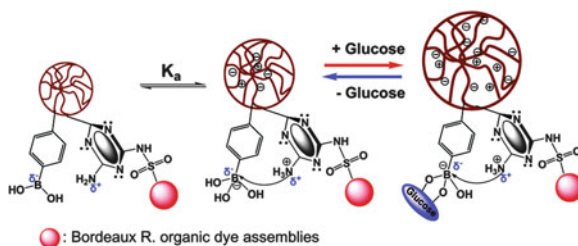


Fig. 16 Schematic illustration of glucose sensing with dye-functionalized boronic acid microgels. Reproduced with permission from Li and Zhou [146]. Copyright 2013 The Royal Society of Chemistry

[141–144]. Related block copolymers (**55**, **56**) were investigated by Kim and coworkers as building blocks of responsive vesicles for insulin release applications [36, 40]. Narain and coworkers found that the combination of boroxole-functionalized polyacrylamide copolymer **57** with glycopolymers resulted in temperature-, pH-, and glucose-responsive gels [39]. Finally, Zhou et al. developed microgels that allow for glucose sensing at physiological pH (due to the presence of quaternizable amino or pyridyl groups) and at the same time contain organic dyes as reporter molecules [145]. Glucose binding resulted in swelling of the gel, which affected the dye–dye separation and in turn triggered a change in the photoluminescence properties (Fig. 16) [146].

Boronic and borinic acid-functionalized polymers have also attracted interest as new responsive and smart materials. Sumerlin and coworkers reported that the cross-linking of a boronic acid-functionalized acrylamide block copolymer (**58**) with polyfunctional alcohols leads to formation of star-shaped polymer architectures as illustrated in Fig. 17a [147]. Addition of monofunctional diols triggered disassembly into the building blocks.

We recently introduced a new type of borinic acid polymer (**59**) that is capable of hydrogen bonding with Lewis bases [38] In DMSO solution, the polymer shows an upper critical solution temperature (UCST) that strongly depends on the amount of water present (Fig. 17b). An essentially linear correlation of the UCST on the amount of water was found over an unusually wide temperature range. The solubility characteristics were also reversibly altered by the addition of a small amount of fluoride anions.

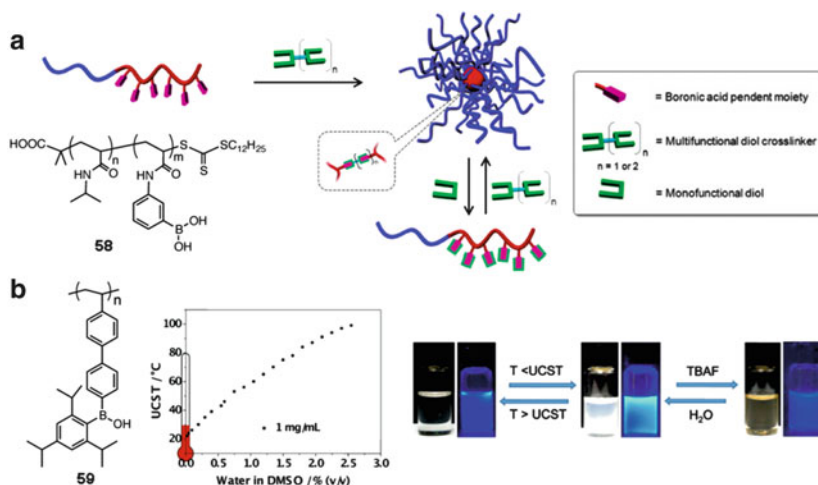


Fig. 17 (a) Schematic illustration of the reversible cross-linking of **58** with multifunctional diols to give starlike assemblies. Adapted with permission from Bapat et al. [147]. Copyright 2011 American Chemical Society. (b) Effect of water content on the UCST of **59** in DMSO solution and photographs (visible light/UV irradiation) illustrating changes with temperature and upon addition of fluoride anions. Adapted with permission from Wan et al. [38]. Copyright 2014 Wiley-VCH Verlag GmbH & Co. KGaA

3.4 Borate and Boronium-Functionalized Polymers as Polyelectrolytes and Polymeric Ligand

Organoborate polymers are attractive as catalyst supports and as new materials for lithium ion battery applications [10]. Typical structural motifs are shown in Fig. 18. For example, Ohno, Matsumi, and coworkers reacted lithium mesitylhydroborate with poly(ethylene oxide) (PEO) and then added different alcohols (ROH) to give polymers **60**. The organoborate moieties in the main chain act as weakly coordinating counterions to lithium ions [148].

The combination of imidazolium ions and borate functionalities in the zwitterionic polymer **61** was found to be beneficial as it resulted in high conductivity despite a relatively high glass transition temperature [149]. In polymer **62**, the amido group is stabilized by B–N π – π interactions, which leads to high ionic conductivity due to favorable ion dissociation. In recent years new composite materials, for example with cellulose, have also been developed [150].

Organoborate polymers are also advantageous as weakly coordinating counterions in catalysis. Many transition metal catalysts rely on the presence of coordinatively unsaturated cationic metal sites, which are stabilized by weakly coordinating counteranions [151, 152]. Especially fluorinated organoborate derivatives are highly effective [153]. Efforts at incorporating these weakly coordinating borate anions into polymeric materials resulted in a range of new ionic polymers [154–156]. Mecking, Vogt, and coworkers prepared borate-functionalized polymer

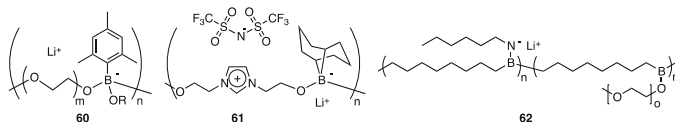


Fig. 18 Examples of boron-containing polymers as electrolytes for lithium ion batteries

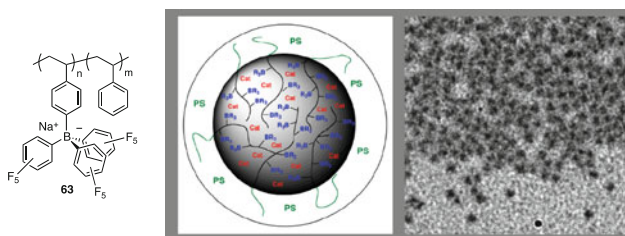


Fig. 19 Self-assembly of the amphiphilic block copolymer **63** in toluene solution. The TEM micrograph shows micelles that had been loaded with $[\text{Rh}(\text{COD})(\text{dppb})]^+$. Adapted with permission from Cui et al. [156]. Copyright 2010 American Chemical Society

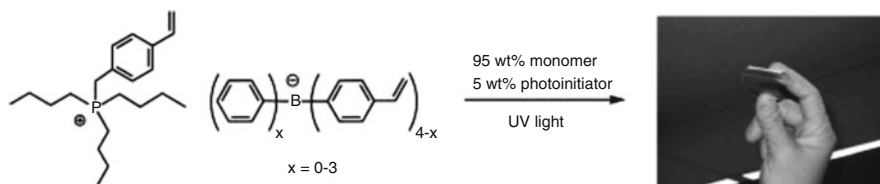


Fig. 20 Formation of highly cross-linked phosphonium borate films by photopolymerization of multifunctional monomers. Adapted with permission from Berven et al. [157]. Copyright 2013 Wiley-VCH Verlag GmbH & Co. KGaA

particles by emulsion copolymerization of styryltriphenylborate, styrene, and divinylbenzene and used the particles to immobilize $[\text{Rh}(\text{dppp})(\text{COD})]^+$. The immobilized catalyst was then applied in the hydrogenation of α -acetamidocinnamic acid [155]. We prepared polystyrene-based homo and block copolymers with perfluoroarylborate pendant groups using a polymer modification protocol [156]. In toluene, the block copolymer **63** assembled into micelles that contain the functional borate moieties in the core surrounded by an inert polystyrene shell (Fig. 19). These micelles were loaded with the $[\text{Rh}(\text{COD})(\text{dppb})]^+$ complex.

Ragogna and coworkers reported an innovative approach to cross-linked borate materials. Highly cross-linked films were generated by photoinitiated polymerization of styryl-functionalized phosphonium borate salts (Fig. 20) [157]. With an increasing number of styrene moieties in the phosphonium borates, the polymer films became harder and more scratch resistant, indicating the tunability of the material properties by simple structural modifications.

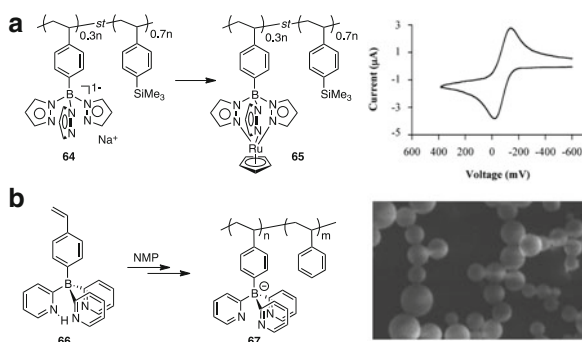


Fig. 21 (a) Metal complexation of a Tp-functionalized polymer **64** and cyclic voltammetry data of the product **65**. Adapted with permission from Qin et al. [35]. Copyright 2008 American Chemical Society. (b) Synthesis of Tpyb-functionalized block copolymer **67** and SEM picture of self-assembled particles after treatment with Cu²⁺ ions. Adapted with permission from Shipman et al. [54]. Copyright 2013 American Chemical Society

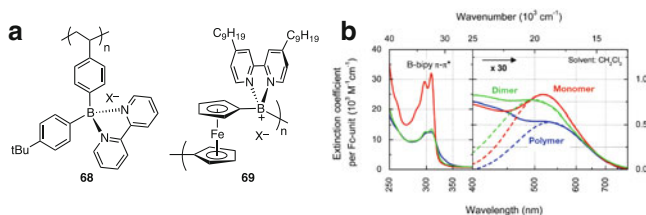


Fig. 22 (a) Structures of boronium-functionalized polymers. (b) UV-visible spectra of polymer **69** in comparison to the corresponding monomer and dimer. Adapted with permission from Cui et al. [161]. Copyright 2010 American Chemical Society

The organoborate polymers discussed thus far are weakly coordinating. Another approach is to attach strong binding sites to generate powerful polymeric ligands for applications in catalysis and supramolecular materials chemistry. We anchored tris (1-pyrazolyl)borate (Tp) [158] ligands to polystyrene by polymer modification processes and then complexed the polymer (**64**) with organoruthenium moieties [35]. Cyclic voltammetry data on **65** confirmed the metal attachment and revealed excellent reversibility for the metal oxidation process (Fig. 21a). We also investigated related Tp end-functionalized polymers as building blocks for supramolecular polymers [159]. More recently, we introduced a new class of tris (2-pyridyl)borate (Tpyb) ligands [160]. Tpyb ligands are more robust than the Tp ligands, allowing for direct polymerization of the styryl derivative **66** by NMP (Fig. 21b) [54]. The block copolymer **67** was generated by sequential polymerization and its nanostructures and metal complexation were studied.

In comparison to these anionic borate-functionalized polymers, cationic organoboron ionomers and polyelectrolytes remain quite rare. The boronium cation sites in polymers **68** (Fig. 22a) are isoelectronic to ammonium functionalities. They were introduced by simple “click”-like reaction of the corresponding bromoborane-

functionalized polymers with 2,2'-bipyridine. Essentially quantitative functionalization was achieved [161–163]. Corresponding amphiphilic block copolymers with polystyrene as a second block were also prepared and their micellar assembly was studied in MeOH. Potential utility as antibacterial materials similar to ammonium-functionalized polymers was suggested. Boronium cation-bridged polyferrocenes (**69**) were introduced as redox-active polyelectrolytes. The ferrocene moieties undergo reversible oxidation at moderate potentials, while the bipyridylboronium heterocycles can be reversibly reduced. These materials display an intense purple color in solution due to charge transfer between the electron-rich ferrocene and the electron-poor bipyridylboronium groups (Fig. 22b).

4 Conclusions

As this brief review indicates, the much greater sophistication of today's synthetic toolbox for borane incorporation into polymeric materials has enabled tremendous progress in terms of (1) the types of functionalities that can be incorporated; (2) the architectural precision that can be achieved; and (3) the new applications that are under pursuit as a consequence. The future of borane chemistry in polymer science is certain to be bright!

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